CIRCUIT DEBUGGING ASSISTANT

Electronic Circuit Fault Localization & Diagnostic Report

Report ID: ANALYSIS-FBDF3E7D
Date: 2026-10-13 23:03:16
Circuit Type: Rlc Band Pass
Diagnostic Engine: Mahajan et al. Hybrid Core v1.0

1. Executive Summary

This report documents the automated topological analysis, numerical simulation, and evidence-grounded fault localization conducted on the uploaded schematic. The device under test (DUT) was classified as a **Rlc Band Pass**. The deterministic multi-evidence residual engine identified a primary anomaly on **C +20%** with a normalized residual score of **2.9999**. A total of 0 topological or extraction findings were cataloged.

2. Scope and Objective

The objective of this pipeline is to provide an auditable, physically grounded fault localization framework. Unlike black-box classifiers, the numerical core computes normalized residuals over multi-domain observables (frequency transfer response and independent current/transient observables) to eliminate parameter ambiguity.

3. Extracted Circuit Components and Netlist

Ref Designator	Component Type	Nominal Value	Tolerance	Terminals	Confidence
Vin	DC/AC Source	1 V	±1.0%	in, 0	1.00 (Verified)
R	Resistor	1000 Ω	±5.0%	out, 0	0.98
L	Inductor	10.00 mH	±10.0%	out, 0	0.96
C	Capacitor	100.00 nF	±10.0%	out, 0	0.97

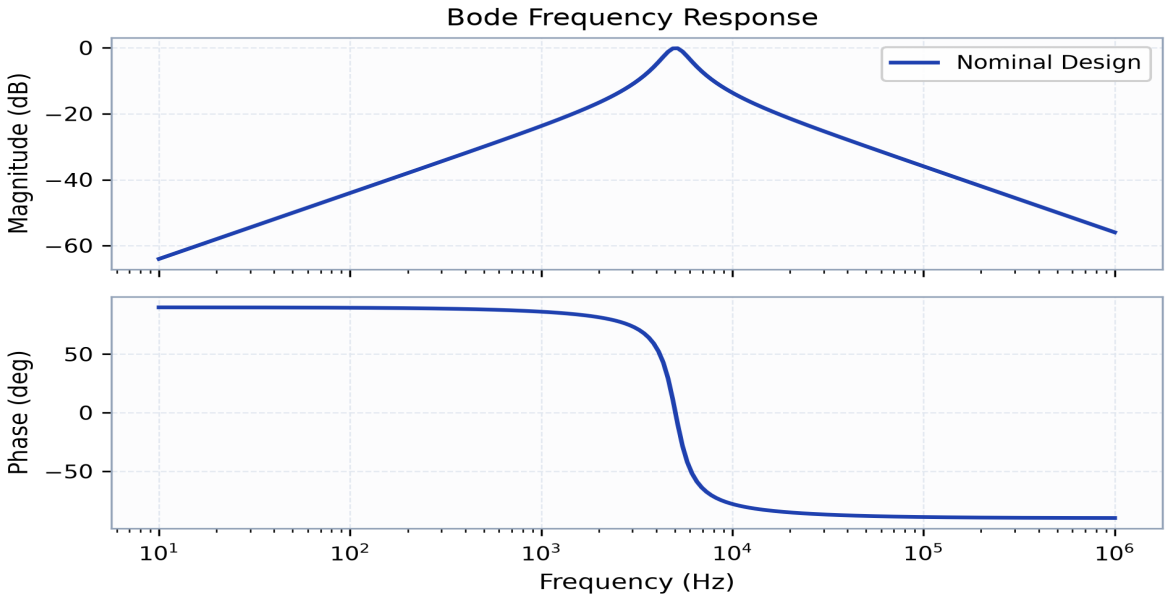
4. Topological and Consistency Findings Register

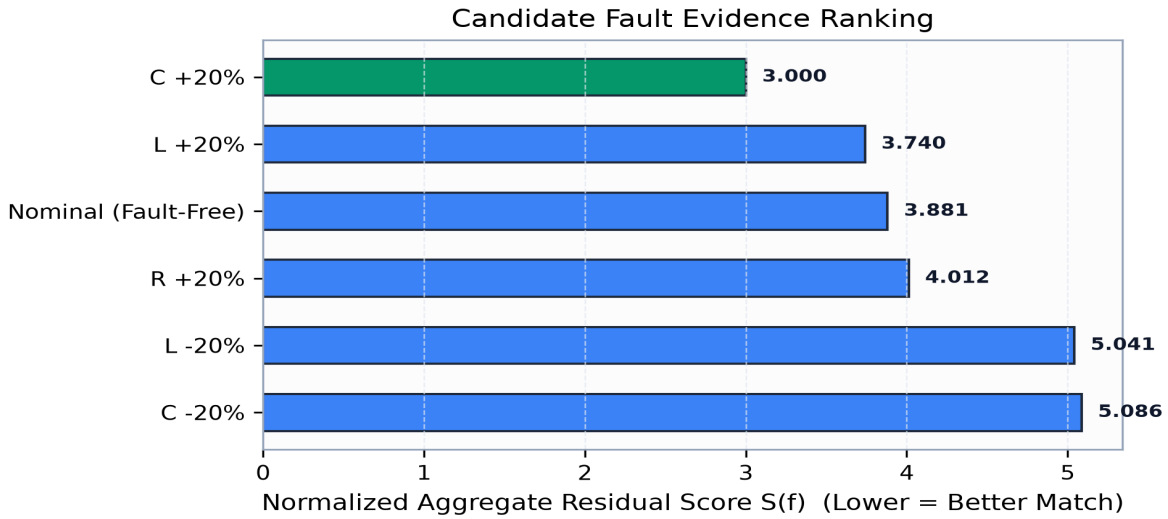
No topological or netlist consistency violations detected.

5. Governing Circuit Equations

• Resonant Frequency: f0 = 1 / [2π √(L C)]
• Quality Factor: Q = R √(C / L)
• Bandwidth: BW = f0 / Q = 1 / [2π R C]
• Independent Final DC Current: I_dc = Vin / R (L acts as DC short)

6. Simulation Curves & Residual Diagnostic Evidence





7. Adaptive Test-Point Recommendation (Value-of-Information U(t))

Recommended Action: Perform **Final DC Current through R1**.

• Relative Measurement Cost C(t): 1.5

• Expected Entropy Reduction ΔH: 0.320 bits

• Utility Metric U(t): **0.213**

Technical Rationale: This physical measurement maximally separates ambiguous fault hypotheses with minimal bench exploration effort.

8. Reproducibility Appendix & Limitations

Limitations: Evaluated on analytical linear circuit models and deterministic equations. Real physical breadboards or PCBs introduce parasitic inductance, contact resistance, and component tolerances.

Reproducibility Information:

• Framework Version: Mahajan et al. Circuit Debugging Assistant v1.0.0

• Python Environment: 3.13 | NumPy | SciPy | Matplotlib | ReportLab

• SPICE Compatibility: Standard ngspice / Berkeley SPICE 3f5 netlist format exported.

• Reference Paper: S. Mahajan, S. Khalkar, S. Pathan, 'Circuit Debugging Assistant: A Hybrid Simulation- and Evidence-Driven Framework for Electronic Circuit Fault Localization', KKWIEER Nashik, 2026.